

NON-MF GROUPS

SAUERS

We show that not every countable group is MF, where MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. The main tool is a one-sided compression criterion that places normal Kazhdan subgroups in the MF radical. Applied to the binary Leavitt group, it shows that the finitely generated simple property-(T) group $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ has trivial image under every homomorphism to an MF group; consequently its reduced group C^* -algebra is a separable stably finite C^* -algebra that is not MF. A second application gives a finitely presented torsion-free group with property (T) that has trivial image under every homomorphism to an MF group, and every nontrivial quotient of it has the same property.

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1. INTRODUCTION

Not every countable group is MF. Our counterexample is

$$H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2)).$$

The group H is finitely generated, nontrivial, simple, and has property (T), but every homomorphism from H to an MF group is trivial. In particular, H is not MF. Its reduced group C^* -algebra $C_r^*(H)$ is separable and stably finite, but is not MF.

More precisely, a countable group G is *MF* if there are positive integers d_n and maps $V_n: G \rightarrow \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE]. Here, MF group always means this operator-norm local-approximation property. The strong-convergence convention also

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requires the matrix models to recover the norms of the left regular representation [GKM, Sch].

Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \rightarrow 0\}$$

is the c_0 -direct sum.

A separable C^* -algebra is called MF if it embeds as a C^* -subalgebra of a norm matrix corona. For every countable group G , the reduced group algebra $C_r^*(G)$ is separable and has a faithful canonical trace, hence is stably finite. A non-MF group therefore gives a separable stably finite reduced group C^* -algebra that is not MF (see the proof of Theorem B).

For a countable group G , define its MF radical by

$$\text{Rad}_{\text{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

We call a homomorphism $G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. Proposition 2.1 shows that G is MF precisely when $\text{Rad}_{\text{MF}}(G)$ is trivial. If $\text{Rad}_{\text{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

For a subgroup $L \leq G$, put

$$\text{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression-centralizer defect* by

$$(1) \quad \mathfrak{D}_G(L) = \langle\langle [ucu^{-1}, \ell] : u \in \text{Comp}_G(L), c \in C_G(L), \ell \in L \rangle\rangle_G.$$

Theorem A (one-sided compression criterion). *Let G be countable and let $L \leq G$ have property (T). If $K \trianglelefteq G$ has property (T) and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) and $\mathfrak{D}_G(L) = G$, then $\text{Rad}_{\text{MF}}(G) = G$.

We sketch the argument. Corollary 2.6 gives $\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G)$. That is, every operator norm asymptotic representation of G is asymptotically trivial on $\mathfrak{D}_G(L)$ in normalized Hilbert–Schmidt norm. This holds because conjugation by the compressor moves the Kazhdan projection of L below itself, and stable finiteness of the corona forces equality (Section 2.3). If a corona homomorphism were nontrivial on K , the Kazhdan projection of K would cut out a G -invariant corner on which K has no nonzero fixed vectors (Lemma 2.7); by the Kazhdan inequality some $s_0 \in K$ would then stay bounded away from the identity in normalized Hilbert–Schmidt norm along a subsequence, contradicting $s_0 \in \mathfrak{D}_G(L)$.

Theorem 2.2 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of G over any field kills $\mathfrak{D}_G(L)$.

Property (T) is not itself an obstruction to MF approximation: every residually finite group is MF, including many Kazhdan groups. In the argument above, property (T) controls the fixed vectors of an exact corona representation. The one-sided compression forces the corresponding Kazhdan projection to commute with the compressor, which makes the Hilbert–Schmidt commutant invariant.

We next apply Theorem A to the group H .

Let $L_{\mathbb{F}_2}(1, 2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$ -algebra generated by s_0, s_1, t_0, t_1 subject to

$$(2) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring R , write $e_{ij}(a) = 1 + aE_{ij}$ and let $\text{EL}_n(R)$ be the subgroup of $\text{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group). *Put $R = L_{\mathbb{F}_2}(1, 2)$ and $H = \text{EL}_{12}(R)$. Then H is finitely generated, nontrivial, simple, has property (T), and*

$$\text{Rad}_{\text{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF. Its reduced group C^ -algebra $C_r^*(H)$ is separable and stably finite, but is not MF.*

Inside H , an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the subgroup $L \cong \text{EL}_3(R)$ in the upper left. The element $c = e_{34}(1)$ centralizes L , and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation $t_1(s_1 t_1)s_1 = 1$ and the Steinberg commutator relations show directly that d normally generates H . Since $d \in \mathfrak{D}_H(L)$ and $\mathfrak{D}_H(L)$ is normal, it follows that $\mathfrak{D}_H(L) = H$. Therefore Theorem A gives $\text{Rad}_{\text{MF}}(H) = H$.

The second application of Theorem A is to a group without torsion.

Theorem C (a torsion-free group with full MF radical). *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property (T) and*

$$\text{Rad}_{\text{MF}}(Q) = Q.$$

Every nontrivial quotient of Q also equals its own MF radical; in particular, no nontrivial quotient of Q is MF.

The group Q is a small-cancellation quotient, in the sense of Hull [Hu], of the torsion-free property-(T) group of Fournier-Facio [FF, §2], chosen so that the compression defect normally generates the quotient (Section 4). Thus the obstruction of Theorem A depends neither on torsion, nor on simplicity, nor on the Leavitt algebra.

Relation to prior work. Blackadar and Kirchberg developed the notions of MF and NF algebras and proved that a separable C^* -algebra is NF if and only if it is nuclear and MF [BK]. It also asked whether every separable nuclear stably finite C^* -algebra is quasidiagonal. This was a question about nuclear C^* -algebras, not about arbitrary groups.

The group property used here was introduced by Carrión–Dadarlat–Eckhardt [CDE]. If every separable stably finite C^* -algebra were MF, then every countable group would be MF because its reduced group C^* -algebra is separable and stably finite. Later papers on approximation of groups relate the resulting operator norm problem to Kirchberg’s question [LO, Th]. The negative solution of the Connes embedding problem [JNVWY] yields separable stably finite C^* -algebras that are not MF, as Goldbring and Hart observe [GH, Proposition 6.1 and Remark 6.2]. Theorem B gives such an example among reduced group C^* -algebras. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves that $A *_C A$ is MF whenever A is a separable MF C^* -algebra and $C \subseteq A$ is a C^* -subalgebra [Sh, Theorem 10]. She also proves that the full group C^* -algebras of amalgamated free products of amenable groups and of certain semidirect products are MF [Sh]. Nonapproximability was known for the unnormalized Schatten p -norms with $1 < p < \infty$ [DGLT, LO]. Bachner–Dogon–Lubotzky later proved nonapproximability for $p = 1$ [BDL]. The operator norm case $p = \infty$ remained open. They also identify stability from operator norm to normalized Hilbert–Schmidt norm as a possible route to a non-MF group [BDL, Proposition 1.5]; that is the route taken here.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. The compression used here comes from their defining relations. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser classified the subgroups of GL_n over an exchange ring that are normalized by EL_n [Pre]. His sandwich theorem, together with the simplicity of the binary Leavitt algebra and the equality $Z(R) = \mathbb{F}_2$, implies that $EL_{12}(R)$ is simple (Proposition 3.3). Ershov–Jaikin–Zapirain give property (T) for $EL_n(R)$ whenever $n \geq 3$ and R is a finitely generated unital associative ring [EJZ, Theorem 1.1]. OpenAI used the binary Leavitt unit group to construct the first known nonsofic group [OAI, Chapter 3]. Khanh–Thanh’s matrix-ring isomorphisms and their equality $GL_n(R) = EL_n(R)$ identify that group with the group H studied here [K–T, Proposition 4.2 and Corollary 4.4]. We apply the Leavitt compression to operator norm approximation. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. ONE-SIDED COMPRESSION AND THE MF RADICAL

2.1. The MF radical. The image of a corona homomorphism from G is a countable subgroup of a corona unitary group, hence MF; conversely, every

countable MF group embeds in the unitary group of a norm matrix corona. Thus $\text{Rad}_{\text{MF}}(G)$ is also the intersection of the kernels of all homomorphisms from G to MF groups.

Proposition 2.1 (basic properties of the MF radical). *For every countable group G , the subgroup $\text{Rad}_{\text{MF}}(G)$ is fully invariant, the quotient $G/\text{Rad}_{\text{MF}}(G)$ is MF, and G is MF if and only if $\text{Rad}_{\text{MF}}(G)$ is trivial.*

Proof. An intersection of kernels is normal. If α is an endomorphism of G , $x \in \text{Rad}_{\text{MF}}(G)$, and π is a corona homomorphism from G , then $\pi \circ \alpha$ kills x . Hence $\pi(\alpha(x)) = 1$ for every π , which proves full invariance.

Put $R = \text{Rad}_{\text{MF}}(G)$. The trivial group is MF. If G/R is nontrivial, let $x \in G/R$ be nontrivial and choose $g \in G$ mapping to x . Then $g \notin R$. Some homomorphism from G to an MF group therefore separates g from the identity. It kills R and descends to a homomorphism from G/R that separates x . Thus G/R is residually MF, and hence MF by Korchagin [Kor, Proposition 6].

If G is MF, a corona embedding of G has trivial kernel, so $\text{Rad}_{\text{MF}}(G) = 1$. Conversely, if $\text{Rad}_{\text{MF}}(G) = 1$, then $G = G/\text{Rad}_{\text{MF}}(G)$ is MF. $\square$

2.2. Finite dimension.

Theorem 2.2 (finite-dimensional commutant rigidity). *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k -vector space, and $\rho: G \rightarrow \text{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\text{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned} \rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u) \\ &= \rho(u)^{-1}x\rho(h')\rho(u) \\ &= \rho(u)^{-1}x\rho(u)\rho(h). \end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. Therefore

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

$\square$

Finite dimensionality is essential: an injective linear self-map of an infinite-dimensional commutant need not be surjective.

2.3. Kazhdan transport in normalized Hilbert–Schmidt norm. For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n: G \rightarrow \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \rightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G . Define

$$(3) \quad R_{\infty \rightarrow 2}(G) = \bigcap_V K_2(V),$$

where V ranges over all operator norm asymptotic representations of G .

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \left| \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \rightarrow 0 \quad (\ell \in L) \end{array} \right. \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\text{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.3. *For every sequence of positive integers (m_n) , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v . Then $x_n^*x_n \rightarrow 1$ in norm. For all sufficiently large n , the matrix x_n is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If A is a stably finite unital C^* -algebra, equivalent projections $p \leq q$ satisfy $p = q$. Indeed, if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1 - q)$ is an isometry in A . Consequently $ww^* = q$, and hence $p = q$. $\square$

Lemma 2.4 (one-sided order for the Kazhdan projection). *Let L have property (T), let B be a unital C^* -algebra, and let $\pi: L \rightarrow \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C_{\max}^*(L) \rightarrow B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \leq P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\text{Fix } \pi(L)$. If ξ is L -fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^*\text{Fix } \pi(L) \subseteq \text{Fix } \pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B . $\square$

Theorem 2.5 (one-sided Kazhdan transport). *Let $L \leq G$ have property (T), and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator norm asymptotic representation of G . Then*

$$\text{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\text{Ad}(A) - \text{Ad}(B)\| \leq 2\|A - B\|,$$

so the conjugation actions are multiplicative modulo c_0 because the multiplicative defects of (V_n) tend to zero in operator norm. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\tilde{\sigma}(g)$ be the class of the coordinate operators $\text{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2 . By the preceding estimate, $g \mapsto \tilde{\sigma}(g)$ is a homomorphism $G \rightarrow \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$ -homomorphism $C_{\max}^*(L) \rightarrow \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\text{Fix } \sigma(L)$. The projection identity below is proved inside $\mathcal{B}$; the representation of $\mathcal{B}$ on $\mathcal{K}_\omega$ is used only afterwards. Since $uLu^{-1} \leq L$, the unitary $U = \tilde{\sigma}(u)$ satisfies the hypothesis of Lemma 2.4, and hence $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 2.3 therefore gives $U^*PU = P$, so U commutes with P . Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\text{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. $\square$

Corollary 2.6. *Let $L \leq G$ have property (T), let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G) \quad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 2.5, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \longrightarrow 0.$$

Therefore

$$\|V_n([ucu^{-1}, \ell]) - 1\|_2 \longrightarrow 0 \quad (\ell \in L).$$

Intersecting over (V_n) proves the claim. $\square$

2.4. Normal Kazhdan subgroups and the MF radical. Let $K \trianglelefteq G$ have property (T), and let p be the image of its Kazhdan projection under a corona homomorphism of G . Normality of K implies that p commutes with the image of G . Hence $q = 1 - p$ also commutes with that image, and in every nondegenerate representation of the corner $q\mathcal{Q}_{\mathbf{d}}q$ the induced representation of K has no nonzero fixed vectors.

Lemma 2.7 (central corona corners). *Let $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \text{rank}(q_n)$, unitary identifications $J_n: \mathbb{C}^{r_n} \rightarrow q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n: G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$. In the corona over the retained coordinates, the class of $(J_n W_n(g) J_n^)$ is the coordinate restriction of $q\rho(g)$.*

Proof. Lift q to projections q_n , and lift each $\rho(g)$ to unitaries $U_n(g)$, choosing $U_n(1) = I_{d_n}$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \quad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n \mathbb{C}^{d_n}$, has both unitarity defects converging to zero. For each fixed g , let $\widetilde{W}_n(g)$ be its polar correction whenever both defects are at most $1/2$, and set $\widetilde{W}_n(g) = q_n$ at the finitely many earlier stages. Then $\widetilde{W}_n(g)$ is unitary in the corner and satisfies

$$\|\widetilde{W}_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

For $g = 1$, the compression is already the identity q_n of the corner, so take $\widetilde{W}_n(1) = q_n$. Since $q \neq 0$, infinitely many q_n are nonzero. Retain and relabel

those coordinates. Put $r_n = \text{rank}(q_n)$, choose a unitary $J_n: \mathbb{C}^{r_n} \rightarrow q_n \mathbb{C}^{d_n}$, and define

$$W_n(g) = J_n^* \widetilde{W}_n(g) J_n \in \mathcal{U}(r_n).$$

Conjugation by J_n preserves operator norms and multiplicative defects. Thus (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$. The displayed estimate identifies the class of $(J_n W_n(g) J_n^*)$ with the coordinate restriction of $q\rho(g)$. $\square$

Theorem 2.8 (normal Kazhdan radical theorem). *Let G be countable and let $K \trianglelefteq G$ have property (T) with $K \leq R_{\infty \rightarrow 2}(G)$. Then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K . Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ . The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. It is contained in $R_{\infty \rightarrow 2}(\Theta(G))$, because an operator norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map, is one of the original group.

Let $p_K \in C_{\max}^*(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K -fixed vectors. Normality of K gives

$$\Theta(g)p\Theta(g)^* = p \quad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g) \text{Fix } \Theta(K)$ and $\text{Fix } \Theta(K)$; these agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero: if $p = 1$, then every vector is fixed by K , so every element of K has trivial corona image. Lemma 2.7 gives positive integers r_n and an operator norm asymptotic representation $W_n: G \rightarrow \mathcal{U}(r_n)$. Let $\widehat{\Theta}: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{r}})$ be its corona homomorphism. Under the coordinatewise corner identifications, $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$.

Choose a finite Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. Represent $\mathcal{Q}_{\mathbf{r}}$ faithfully. The Kazhdan projection of K under $\widehat{\Theta}$ is zero, so $\widehat{\Theta}|_K$ has no nonzero fixed vectors, and the Kazhdan inequality gives, for the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (\widehat{\Theta}(s) - 1)^* (\widehat{\Theta}(s) - 1)$$

the inequality

$$b \geq \frac{\kappa^2}{|S|} 1.$$

The coordinate elements

$$b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - I_{r_n})^* (W_n(s) - I_{r_n})$$

represent b . Taking the normalized trace on $M_{r_n}(\mathbb{C})$ yields

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)I_{r_n}$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from I_{r_n} . This contradicts $s_0 \in K \leq R_{\infty \rightarrow 2}(G)$, established after passing to $\Theta(G)$. Thus every corona homomorphism kills K , as required. $\square$

Proof of Theorem A. For every $u \in \text{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 2.6 gives

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G).$$

Normality of $R_{\infty \rightarrow 2}(G)$ therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G).$$

Since $K \leq \mathfrak{D}_G(L)$, Theorem 2.8 now gives $K \leq \text{Rad}_{\text{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\text{Rad}_{\text{MF}}(G) = G$. $\square$

By Theorem 2.2, every finite-dimensional linear representation of G kills the generators of $\mathfrak{D}_G(L)$, and its kernel is normal, so it kills $\mathfrak{D}_G(L)$. Consequently, $\mathfrak{D}_G(L) = 1$ if G has a faithful finite-dimensional linear representation. The same conclusion holds if G is residually finite, because every nonidentity element then survives in a finite permutation representation. If G is amenable, its property-(T) subgroup L is finite [BHV]; then $uLu^{-1} = L$, so ucu^{-1} centralizes L for every $u \in \text{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 2.5.

3. THE BINARY LEAVITT GROUP

The relation $s_0t_0 + s_1t_1 = 1$ of the binary Leavitt algebra gives an element $\tau \in H$ with $\tau L \tau^{-1} \leq L$ for a copy L of $\text{EL}_3(R)$, and a commutator $d \neq 1$ in $\mathfrak{D}_H(L)$; since d normally generates H , $\mathfrak{D}_H(L) = H$, and Theorem A applies.

3.1. The self-compression. Throughout this section $R = L_{\mathbb{F}_2}(1, 2)$ and

$$p = s_0t_0, \quad q = s_1t_1.$$

The relations (2) give

$$(4) \quad p + q = 1, \quad t_1qs_1 = 1.$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices. We identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. Define a map $\Psi: M_3(R) \rightarrow M_3(R)$ by

$$(5) \quad \Psi(A) = qI_3 + s_0At_0.$$

Here s_0At_0 denotes the matrix $(s_0a_{ij}t_0)_{ij}$. The relations $q^2 = q$, $qs_0 = t_0q = 0$, and $t_0s_0 = 1$, all consequences of (2), show that Ψ is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0I_3t_0 = qI_3 + pI_3 = (p+q)I_3 = I_3,$$

so Ψ preserves the identity. Finally, $t_0\Psi(A)s_0 = A$, so Ψ is injective. Hence Ψ restricts to an injective group homomorphism $\mathrm{GL}_3(R) \rightarrow \mathrm{GL}_3(R)$.

To realize Ψ by conjugation on the embedded copy of $\mathrm{GL}_3(R)$, define the following 6×6 block matrices:

$$X = \begin{pmatrix} s_0I_3 & s_1t_0I_3 \\ 0 & t_1I_3 \end{pmatrix}, \quad Y = \begin{pmatrix} t_0I_3 & 0 \\ s_0t_1I_3 & s_1I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $Y = X^{-1}$. Put

$$(6) \quad \tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R).$$

A direct block multiplication gives

$$(7) \quad \tau \mathrm{diag}(A, I_9) \tau^{-1} = \mathrm{diag}(\Psi(A), I_9).$$

Lemma 3.1. *The matrix τ defined in (6) belongs to $\mathrm{EL}_{12}(R)$.*

Proof. For $B \in M_6(R)$, the block matrices $\begin{pmatrix} I & B \\ 0 & I \end{pmatrix}$ and $\begin{pmatrix} I & 0 \\ B & I \end{pmatrix}$ lie in $\mathrm{EL}_{12}(R)$: they are the products of the elementary matrices $e_{r,6+s}(b_{rs})$, respectively $e_{6+r,s}(b_{rs})$, and no cross terms arise because distinct off-diagonal matrix units have product zero. Whitehead's identity

$$\mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}$$

with $I = I_6$ therefore expresses τ as a product of six matrices of $\mathrm{EL}_{12}(R)$. $\square$

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \leq H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

Proposition 3.2. *The groups H and L have property (T), and $\tau \in H$ satisfies*

$$(8) \quad \tau L \tau^{-1} \leq L.$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$.

For $i \neq j$ and $a \in R$, equations (7) and (5) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0at_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of L . These matrices generate L , so (8) follows. $\square$

3.2. Simplicity and the full MF radical.

Proposition 3.3. *The group $H = \text{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], and therefore is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $H = \text{EL}_{12}(R)$, the subgroup $N \leq \text{GL}_{12}(R)$ is normalized by $\text{EL}_{12}(R)$. Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal $I \trianglelefteq R$ such that

$$\text{EL}_{12}(R, I) \leq N \leq C_{12}(R, I).$$

Simplicity of R gives $I = 0$ or $I = R$. If $I = R$, then $\text{EL}_{12}(R, I) = H$, so $N = H$.

Suppose that $I = 0$. Then $N \leq C_{12}(R, 0) = Z(\text{GL}_{12}(R))$. A matrix that commutes with every $e_{ij}(1)$ is a scalar matrix λI_{12} . Commutation with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^\times$. Since $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], we have $\lambda = 1$ and $N = 1$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. $\square$

Proposition 3.4. *Let*

$$c = e_{34}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

in H . Then

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad \langle\langle d \rangle\rangle_H = H.$$

Proof. For $i, j \in \{0, 1, 2\}$ with $i \neq j$ and $a \in R$, we have $j \neq 3$ and $i \neq 4$. Hence the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate L , so $c = e_{34}(1)$ belongs to $C_H(L)$.

Direct multiplication of the matrices in (6) gives

$$(9) \quad \tau c \tau^{-1} = e_{01}(q) e_{34}(1).$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ therefore gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$.

Let $N = \langle\langle d \rangle\rangle_H$. Since $t_1 q s_1 = 1$, the Steinberg relations give

$$[d, e_{23}(s_1)] = e_{03}(q s_1) \in N, \quad [e_{10}(t_1), e_{03}(q s_1)] = e_{13}(1) \in N.$$

Conjugation by elementary Weyl matrices gives $e_{ij}(1) \in N$ for every $i \neq j$. For $a \in R$ and distinct i, j, k ,

$$[e_{ik}(a), e_{kj}(1)] = e_{ij}(a) \in N.$$

Thus N contains every elementary generator of H , so $N = H$. $\square$

Proof of Theorem B. Let Y be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. Since $e_{ij}(a + b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ for distinct i, j, k , and since every element of R is a sum of monomials in s_0, s_1, t_0, t_1 , induction on the length of a monomial shows that every $e_{ij}(a)$ lies in $\langle Y \rangle$; there is always a third index because $12 \geq 3$. Thus $H = \langle Y \rangle$ is finitely

generated, and H has property (T) by the Ershov–Jaikin–Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (8), the centrality of c relative to L , and Proposition 3.4 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 3.4 gives $\langle\langle d \rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\text{Rad}_{\text{MF}}(H) = H$. Proposition 3.3 shows that H is nontrivial and simple.

If M is MF and $f: H \rightarrow M$ is a homomorphism, compose f with a faithful corona embedding of M . Since $\text{Rad}_{\text{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion.

Finally, $C_r^*(H)$ is separable because H is countable. Its canonical trace is faithful, so it is stably finite. If an MF embedding $e: C_r^*(H) \rightarrow \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1 - p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$. This contradicts the fact that H is not MF. $\square$

4. A TORSION-FREE GROUP WITH FULL MF RADICAL

A group is acylindrically hyperbolic if it admits a non-elementary acylindrical action on a hyperbolic space [Os]. For such a group G , Hull [Hu, Theorem 3.12] provides a generating set A such that the Cayley graph $\Gamma(G, A)$ is hyperbolic and the action of G on it is acylindrical and non-elementary; a subgroup $N \leq G$ is *suitable* with respect to A if N acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup of G [Hu, Definition 1.4]. We use Hull’s small cancellation theorem [Hu, Theorem 7.1] in the following form.

Theorem 4.1 (Hull’s small cancellation theorem). *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A , and let $t_1, \dots, t_m \in G$ and a finite subset $F \subseteq G$ be given. Then there is a surjective homomorphism $q: G \rightarrow Q$ such that Q is acylindrically hyperbolic, $q|_F$ is injective, $q(t_i) \in q(N)$ for all i , and every element of finite order in Q is the image of an element of the same order in G .*

Moreover the kernel of q is the normal closure of m elements; this is read off from Hull’s proof: the case $m = 1$ adds the single relator $W = t_1^{-1}h_1^{m_1}h_2^{l_1} \cdots h_1^{m_n}h_2^{l_n}$ for suitable $h_1, h_2 \in N$, and the general case is an induction on m .

Lemma 4.2 (saturation). *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $F \subseteq G$ be finite. Then there is a surjective homomorphism $q: G \rightarrow Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q|_F$ is injective, and $q(N) = Q$.*

Proof. Fix A as above. Since G is torsion-free and $N \neq 1$, N is infinite, so $|N \cap gNg^{-1}| = \infty$ for every $g \in G$, and by Osin [Os, Lemma 7.1] N acts non-elementarily on $\Gamma(G, A)$. The only finite subgroup of G is trivial, so

N is suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable. Apply Theorem 4.1 to N_0 with $t_1, \dots, t_m$ a finite generating set of G and with the set F . Then

$$Q = \langle q(t_1), \dots, q(t_m) \rangle \leq q(N_0) \leq q(N) \leq Q,$$

so $Q = q(N_0) = \langle q(h_1), q(h_2) \rangle$ and $q(N) = Q$. The group Q is torsion-free because G is, and finitely presented because it is G with m relators added. $\square$

Following Fournier-Facio [FF, §2], let U be a universal finitely presented torsion-free group, one containing a copy of every finitely presented torsion-free group [Chi]; let S be a finitely presented infinite simple torsion-free group [HL]; and let H_0 be a torsion-free hyperbolic group with property (T) [KK]. Small cancellation over the relatively hyperbolic pair $(U * H_0, U)$ embeds U in a finitely presented torsion-free quotient P of H_0 [FF2, Proposition 2.3]; torsion-freeness is preserved by Osin's embedding theorem [OsSC, Theorem 2.4(5)]. Consequently, P has property (T). By universality, P contains a subgroup $P_1 \times P_2 \times S$ with $P_i \cong P$. Let E be the double HNN extension of P with stable letters u_1, u_2 and $u_i P u_i^{-1} = P_i$; it is finitely presented and torsion-free, and its Bass–Serre action makes it acylindrically hyperbolic by the tree criterion of Minasyan–Osin [MO]. Hull's common quotient theorem [Hu, Corollary 7.4] applied to E and H_0 gives a surjection $\pi: E \rightarrow G_0$ onto a finitely presented torsion-free group with property (T), injective on a prescribed finite subset of E ; since E is finitely generated, the proof of that corollary produces G_0 acylindrically hyperbolic. We run the construction with the prescribed finite subset containing an element $s \neq 1$ of S . Then $\pi(S) \neq 1$, so $\pi|_S$ is injective because S is simple.

Put $\Gamma = \pi(P)$, $t = \pi(u_1)$, and $J = t^{-1}\pi(S)t$. Since S commutes with $P_1 = u_1 P u_1^{-1}$, the subgroup J centralizes Γ , and $t\Gamma t^{-1} = \pi(P_1) \leq \Gamma$, so $t \in \text{Comp}_{G_0}(\Gamma)$. Moreover $tJt^{-1} = \pi(S) \leq \Gamma$, and Γ has property (T) as a quotient of P .

Lemma 4.3. *Let $\rho: G_0 \rightarrow L$ be a surjective homomorphism with $\rho(\pi(S)) \neq 1$. Then $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$.*

Proof. Write $\Gamma_L = \rho(\Gamma)$, $t_L = \rho(t)$, and $J_L = \rho(J)$. Then $t_L \Gamma_L t_L^{-1} \leq \Gamma_L$, J_L centralizes Γ_L , and $t_L J_L t_L^{-1} = \rho(\pi(S)) \leq \Gamma_L$. For $c \in J_L$ and $\ell \in \Gamma_L$, the commutator $[t_L c t_L^{-1}, \ell]$ lies in $\mathfrak{D}_L(\Gamma_L)$ by (1). As c ranges over J_L , the element $t_L c t_L^{-1}$ ranges over $\rho(\pi(S))$, and ℓ may be taken in $\rho(\pi(S)) \leq \Gamma_L$; hence the commutator subgroup of $\rho(\pi(S))$ lies in $\mathfrak{D}_L(\Gamma_L)$. Since $\rho(\pi(S)) \neq 1$ and S is simple, $\rho(\pi(S)) \cong S$ is perfect, so $\rho(\pi(S)) \leq \mathfrak{D}_L(\Gamma_L)$. $\square$

Proof of Theorem C. Let N be the normal closure of $\pi(S)$ in G_0 . It is nontrivial, so Lemma 4.2 applied to G_0 , N , and $F = \{1, \pi(s)\}$ gives $q: G_0 \rightarrow Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q(N) = Q$, and $q(\pi(s)) \neq 1$. The group Q has property (T) as a quotient of G_0 .

Let $r: Q \rightarrow L$ be a surjection onto a nontrivial group and put $\rho = r \circ q$. Since $q(N) = Q$, the normal closure of $\rho(\pi(S))$ in L is $\rho(N) = L$; as $L \neq 1$, this forces $\rho(\pi(S)) \neq 1$. By Lemma 4.3, $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$, and since $\mathfrak{D}_L(\rho(\Gamma))$ is normal in L and contains a normal generating set of L , it equals L . The group L has property (T) as a quotient of Q , and $\rho(\Gamma)$ has property (T) as a quotient of Γ . Theorem A, applied to L with the subgroup $\rho(\Gamma)$ and $K = L$, gives $\text{Rad}_{\text{MF}}(L) = L$. Taking r the identity gives $\text{Rad}_{\text{MF}}(Q) = Q$, and a nontrivial group equal to its own MF radical is not MF. $\square$

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